

Mark schemes

Q1.

$X_D \sim \text{Po}(24)$

- (a) $H_0 : \lambda$ (or mean) = 2 (or 10)
 $H_1 : \lambda$ (or mean) > 2 (or 10)
Both; or equivalent

B1

$P(Y \geq 17) = 1 - P(Y \leq 16)$
Accept $1 - P(Y \leq 17)$

M1

$= 1 - 0.09730 = 0.027$
AWRT

A1

< 0.10 (10%)

$[z = 2.05 \text{ to } 2.38 > 1.2816]$
Comparison of probability with 0.1
Comparison of z with 1.2816 or 1.6449

M1

Thus evidence, at 10% level, of increase
 in mean daily number of requests

F on probability or on z

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A1F

5

- (b) CV of Y is such that $P(Y \geq \text{CV}) \leq 0.10$ (10%)
Can be implied by 13, 14 or 15
Accept $P(Y = \text{CV}) = 0.10$

M1

Thus $P(Y \leq \text{CV} - 1) \geq 0.90$
Can be implied by 13, 14 or 15
Accept $P(Y = \text{CV}) = 0.90$

M1

Thus CV = 15
CAO

A1

(c) Power = $1 - P(\text{Type II error})$

Or equivalent

$$= 1 - P(\text{accept } H_0 \mid H_0 \text{ false})$$

$$= P(\text{accept } H_1 \mid H_1 \text{ true})$$

Stated or implied use

B1

$$\lambda = 5 \times 3 = 15$$

Stated or implied use of Po(15)

B1

Thus power = $P(Y \geq CV)$

Attempt at a probability based on C's CV from (b) and Po(15)

M1

$$= P(Y \geq 15) = 1 - P(Y \leq 14)$$

$$= 1 - 0.4657 = 0.53 \text{ to } 0.54$$

AWFW (0.5343)

A1

4

[12]

Q2.

(a) $H_0 : \lambda = 13$

CAO; OE

B1

$H_1 : \lambda < 13$

CAO; OE

B1

$P(R \leq 10 \mid \text{Po}(13))$

Used or implied

M1

$$= 0.2517$$

AWFW 0.251 to 0.252

A1

Prob of $0.2517 > 0.10$ (10%)

$$z = -0.83 \text{ to } -0.70 > -1.28$$



M1

Thus no evidence, at 10% level, of a reduction in the mean value of R

ft on probability or z

In 'context' and qualified

A1ft

6

- (b) Require $P(R \leq r \mid \text{Po}(13)) \approx 0.10$

Stated or implied

M1

Critical Region is $R \leq 8$ or $R < 9$

Accept $R = 8$

May be scored in (a)

A1

2

- (c) Require $P(\text{accept } H_0 \mid H_0 \text{ false})$

OE; PI

B1

$$= P(R > 8 \mid \text{Po}(6.5))$$

Use of $\text{Po}(6.5)$

M1

$$= 1 - P(R \leq 8 \mid \text{Po}(6.5))$$

m1

$$= 1 - 0.7916$$

$$= 0.208 \text{ to } 0.209$$

AWFW (0.2084)

A1

4

[12]

Q3.

- (a) $H_0: \mu = 48$

$$H_1: \mu > 48$$

(both)

B1

$$\bar{X} \approx N\left(\mu, \frac{\sigma^2}{n}\right)$$

Since n is large,

(Central limit theorem)

$\therefore$ for H_0 true,

$$\bar{X} \sim N\left(48, \frac{64}{100}\right) \sim N\left(48, (0.8)^2\right)$$

for 0.64

B1

$$z = \frac{49.2 - 48}{0.8} = 1.50$$

M1A1

$$Z_{crit} = \pm 1.6449$$

B1

$\therefore$ do not reject H_0

A1ft

Insufficient evidence at the 5% level to.
reject the committee's claim
on their z & critical value

E1ft

(b) $P(\bar{X} < 48 + 0.8 \times 1.6449)$

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M1A1

$$= P\left(Z < \frac{49.316 - 50}{0.8}\right)$$

M1

$$= P(Z < -0.8551)$$

A1

$$\begin{aligned} &= 1 - 0.80511 \\ &= 0.19489 \\ &= 0.195 \end{aligned}$$

AWFW 0.189 to 01.98

A1

5

[12]

Q4.

(a) $H_0 : \mu = 65$

B1

$H_1 : \mu > 65$

B1

2

(b) $\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right) = N(65, 0.64)$
for 0.64

B1

for Normal and 65

B1

2

(c) $P(\text{Type I error}) =$
 $P(H_0 \text{ rejected when } H_0 \text{ true})$
 $= P(\bar{X} > 66.4 \text{ when } \mu = 65)$
 $= P\left(Z > \frac{66.4 - 65}{0.8}\right)$

M1

$= P(Z > 1.75)$

area change

m1

$= 1 - 0.95994$

$= 0.04006$

AWRT 0.040

A1

$\therefore$ significance level of test $\approx 4\%$

ft on Type I error

A1ft

4

(d) $P(\text{Type II error}) =$

$P(H_0 \text{ accepted when } H_0 \text{ false})$

$P(\bar{X} < 66.4 \text{ when } \mu = 67)$

$$= P\left(Z < \frac{66.4 - 67}{0.8}\right)$$

M1

$$= P(Z < -0.75)$$

A1

$$= 1 - \Phi(0.75)$$

$$= 1 - 0.77337$$

m1

$$= 0.22663$$

$$= 0.227$$

AWRT 0.23

A1

4

[12]

Q5.

(a) $H_0 : \mu = 300$
 $H_1 : \mu < 300$

B1

$$Y \sim N(300, 16)$$

$$\bar{Y} \sim N\left(300, \frac{16}{20}\right) \sim N(300, 0.8)$$

For 0.8

B1

$$z = \frac{298.1 - 300}{\sqrt{0.8}} = -2.124$$

awrt -2.12

M1A1

$$z_{\text{crit}} = -2.3263$$

Allow ± 2.3263

B1

accept H_0
 Insufficient evidence at the 1% level to
 support the members' suspicion.
on their z

E1f.t.

6

(b) $\frac{\bar{Y} - 300}{\sqrt{0.8}} \leq -2.3263$

M1

$$\bar{Y} \leq 300 - \sqrt{0.8} \times 2.3263$$

A1

$$\bar{Y} \leq 297.9$$

A1

2

(c) P(Type II error)

$$= P\left(Z > \frac{297.9 - 296.5}{\sqrt{0.8}}\right)$$

M1 and next A1 f.t. on their (b)

$$z = 1.5868$$

$$(1.56 - 1.59)$$

M1f.t.

$$= P(Z > -1.59)$$

A1f.t.

$$= 1 - \Phi(1.59)$$

$$= 1 - 0.94408$$

$$= 0.0559 \text{ (3sf)}$$

Awrt 0.055 to 0.060

A1

3

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[11]

Q6.

(a) $H_0 : \mu = 500$

$$H_1 : \mu < 500$$

B1

1

(b) $\bar{Y} \sim N(500, 0.8^2)$

For Normal and 500

B1

For 0.64

B1

2

(c) (i) $P(\text{Type I error}) = 0.05$

B1

1

$$(ii) \quad \frac{\bar{y} - 500}{0.8} = -1.6449$$

Correct z-value (Accept ± 1.645 or ± 1.64)

B1

$$\bar{y} = 498.68$$

M1

$\therefore$ Critical region for $\bar{y}$ is
 $\bar{y} < 498.68$
(AWFW 498.68 and 498.69) (must be a region)

A1

3

(d) $P(\text{Type II error}) = P(\bar{Y} > 498.68)$
Can be $\Rightarrow$ by what follows or diagram.

M1

$$= P\left(Z > \frac{4.98.68 - 497.5}{0.8}\right)$$

for use of $N(4.97.5, 0.8^2)$

$$\text{for } \frac{\bar{y} - 497.5}{0.8}$$

M1A1f.t.

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$$= P(Z > 1.4801)$$

(AWFW 1.48 to 1.49)

A1f.t.

$$= 1 - \Phi(1.48)$$

for $1 - \Phi(z)$

M1

$$= 1 - 0.93056$$

(AWRT 0.931)

B1

$$= 0.0694$$

AWFW 0.0680 to 0.0702

A1

Q7.

(a) (i) $\bar{X} \sim N\left(40, \frac{0.64}{50}\right) \text{ or } N\left(40, \frac{0.8^2}{50}\right)$
or $N(40, 0.0128)$

for Normal & 40

$\frac{\sigma^2}{n}$
Idea of $\frac{\sigma^2}{n}$, correct

$\frac{\sigma^2}{n}$
seen M1

B1/M1A1

(ii) $P(\text{Type I error}) = 0.05$

B1

4

(b) Acceptance region

for $[-1.96, 1.96]$ {wrong symmetrical M1A0}

M1A1

$= (40 - 1.96 \times 0.113, 40 + 1.96 \times 0.113)$

$\frac{\sigma}{\sqrt{n}} = 0.1131 \text{ (A1)}$

M1A1

$= (39.78, 40.22)$

(On their z)

A1f.t.

5

(c) (i) Accepting H_0 when H_0 incorrect

B1

(ii) $P(\bar{X} < 39.78) = P\left(Z < \frac{39.78 - 40.3}{0.1131}\right)$

$= P(Z < -4.61)$

ft on their z from (b)

(allow -4.60)

M1A1f.t.



$$P(\bar{X} < 40.22) = P\left(Z < \frac{40.22 - 40.3}{0.1131}\right)$$

$$= P(Z < -0.6917)$$

(allow -0.7073)

A1f.t.

$$\therefore P(\text{Type II error}) = 1 - \Phi\lambda(0.69) - \Phi\lambda(-4.61)$$

$$= (1 - 0.75490) - 0$$

$$= 0.2451$$

(allow 0.24 to 0.25)

(allow 1 - 0.76115)

M1

(allow 0.239)

A1

6

AWRT

[15]

Q8.

(a) (i) $\bar{x} = 46.5$ $s = 10.81$

46.5 cao and 10.8 (10.8 to 10.82)

B1

$$H_0: \mu = 40$$

B1

$$H_1: \mu \neq 40$$

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one correct hypothesis – generous

both hypotheses correct – ungenerous

Allow $H_1 \mu > 40$

B1

$$t = \frac{(46.5 - 40)}{\frac{10.81}{\sqrt{8}}} = 1.70$$

s.d.

use of their $\frac{\quad}{\sqrt{8}}$

M1

completely correct method for t ignore sign

ml

1.70 (1.695 to 1.705)

A1

c.v $t_7 \pm 2.365$; 1.70 lies between
7df

B1

± 2.365 so accept H_0 , mean is 40 mins
2.365 – their df – ignore sign
(cv 1.895 for one tail test)

B1ft

correct conclusion – must be compared with t

A1ft

Alternatively Allow confidence interval approach

(ii) $H_0 : \mu = 50$

$H_1 : \mu \neq 50$

both hypotheses correct – ungenerous Allow $H_1 \mu < 50$

B1

$$t = \frac{(46.5 - 50)}{\frac{10.81}{\sqrt{8}}} = 0.916$$

-0.916 (-0.915 to -0.916)

A1

c.v $t_7 \pm 2.365$; -0.916 lies between
(cv 1.895 for one tail test)

± 2.365 so accept H_0 , mean is 50 mins

correct conclusion must be compared with
both tails or lower tail of t

A1ft

12

Alternatively Allow confidence interval approach

N.B. Mark a(ii) as a(i) and a(i) as a(ii) if more favourable

(b) Claim 1. **C** Not true – no null hypothesis
rejected so no Type 1 error made

correct conclusion for correct reason – be
generous for E1 but disallow no or clearly
incorrect reason

E2,1

Claim 2. **B** Possibly true – true if
population mean is equal to neither 40 nor 50

correct conclusion for correct reason – be generous for E1 but disallow no or clearly incorrect reason

E2,1

Claim 3. **A** Definitely true – since mean cannot equal both 40 and 50

correct conclusion for correct reason – be generous for E1 but disallow no or clearly incorrect reason

E2,1

6

[18]

Q9.

(a) $\bar{x} = 37.75$ $s = 4.6928$

37.75(37.7, 37.8) & 4.69 (4.69, 4.7)

B1

$H_0: \mu = 40$

One correct hypothesis - generous

B1

$H_1: \mu < 40$

Both hypotheses correct - ungenerous

B1

$$t = \frac{\frac{37.75}{4.6928}}{\sqrt{12}} = -1.66$$

*use of candidate $\frac{s}{\sqrt{12}}$
 method for t- ignore sign
 Completely correct method for t
 -1.66 (-1.65, -1.67)*

M1 m1 m1 A1

c.v. $t_{11} = 1.796$ accept H_0 , conclude no significant evidence to show mean is less than 40 months

11 df

-1.796 (-1.79, -1.8) ignore sign

Conclusion, must be compared with correct tail of t-distribution

B1 B1 A1

10

(b) $H_0: \mu = 40$

$H_1: \mu < 40$

Both hypotheses correct

B1

$$z = \frac{\frac{39.2 - 40}{4.2}}{\sqrt{160}} = -2.41$$

Method for z

M1

c.v. -1.6449 reject H_0 , significant evidence to show mean is less than 40.

-2.41 ($-2.4, -2.42$) and

-1.6449 ($-1.64, -1.655$) ignore sign

Conclusion, must be compared with correct tail of z or t.

A1 A1ft

4

(c) (i) neither, both 5%
neither
both 5%

B1 E1

(ii) neither - cannot make a Type II error if mean is 40

neither

No chance of type II error

B1 E1

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(iii) (a), smaller sample

(a)

Smaller sample

B1 E1

6

[20]

Examiner reports

Q1.

As stated earlier, much of this question proved too much of a challenge for most candidates. In part (a), many candidates failed to state hypotheses in terms of λ . Also, despite the wording of the question, about 50% of candidates attempted an approximate normal test based on either $\lambda = 2$ or $\lambda = 10$. The first is simply incorrect and the second is at best borderline. In general candidates should not opt for approximate tests for binomial proportions or Poisson means when tables in the supplied booklets are available for exact tests.

Only candidates opting for an exact test in part (a) stood any realistic chance of scoring marks in part (b). Most such candidates had a basic idea as to what was needed but only the very best obtained the correct value of 15. The usual alternative was 14, but 13 and 16 were seen. In part (c), most candidates defined power correctly, usually as $1 - P(\text{accept } H_0 \mid H_0 \text{ false})$. They then attempted to calculate this probability using $\lambda = 15$ and so often scored at least 2 marks. A very small proportion of candidates are to be congratulated on scoring all 17 marks in correctly answering this final question.

Q2.

In part (a), candidates were expected to use an 'exact' test since Table 2 tabulates $P(X \leq x \mid \lambda = 13)$. Those who did often scored full marks whereas those who attempted a normal approximation sometimes introduced further inaccuracies. Only the best candidates, usually those who used an exact test in part (a), scored marks in part (b). The same candidates often then scored well in part (c). Those candidates determined to use a normal approximation at all costs usually opted for $N(6.5, 6.5)$ or even $N(6.5, 13)$, neither of which was accepted.

Q3.

There were many fully correct solutions seen to part (a), however, a minority of candidates failed to state their hypotheses correctly whilst others used $\sigma = 8$ where 0.8 was required. Part (b) was usually only answered well by the most able candidates.

Q4.

The hypotheses were usually stated correctly, although there were a few candidates who used $\bar{x}$ instead of μ or who missed out an attempt at μ , altogether. Part (b) was surprisingly poorly done, with many candidates stating $N(65, 32)$ but then going on to use the correct distribution for $\bar{x}$ in part (c). Many candidates calculated correctly the probability of a Type I error in part (c), but then either failed to give the significance level of the test or simply quoted it incorrectly as 5%. Part (d) was answered better than in previous series but it continued to pose problems for average candidates.

Q5.

This was found to be the most difficult question on the paper. In part (a) it was pleasing to see the correct null and alternative hypotheses stated. This was as far as the majority of

candidates were able to progress. The statement $\bar{Y} \sim N\left(300, \frac{16}{20}\right)$ was rarely seen. Even

when it was seen, candidates proceeded to use the wrong tail, evaluating

$$z = \frac{300 - 298.1}{\frac{4}{\sqrt{20}}}$$

$$z = \frac{298.1 - 300}{\frac{4}{\sqrt{20}}}$$

and using, $z_{crit} = +2.3263$ rather than

$$\frac{4}{\sqrt{20}}$$

and $z_{crit} = -2.3263$.

It was not unusual to see $\bar{Y} \sim N(300, 16)$ or $N(298.1, 0.8)$ or $N(300, 4)$ throughout all parts of this question.

Not surprisingly, therefore, parts (b) and (c) were not well done, except by the very best candidates.

Q6.

In part (a) the null hypothesis, $H_0 : \mu = 500$, and the alternative hypothesis $H_1 : \mu < 500$ were often correctly stated. Most candidates also correctly quoted $\bar{Y} \sim N(500, 0.64)$ in part (b) and then used $\sigma = 0.8$ in the remainder of the question.

In part (c)(i) most candidates answered 5% with some converting this to 0.05. Some candidates thought that 5% is equivalent to 0.5. Many candidates found $\bar{Y} = 498.68$ in part (c)(ii) but did not go on to determine the critical region. Candidates were expected to state $\bar{Y} < 498.68$. Some candidates considered this as a 2-tail test using $z = \pm 1.95$ whilst others thought that $z = \pm 1.65$ was an acceptable approximation for $z = -1.6449$.

Candidates performed well when dealing with the calculation of a Type II error in part (d). Many good solutions were seen, but some candidates made the question more difficult by wrongly using a 2-tail configuration.

Q7.

Candidates found this to be the most difficult question on the paper. In part (a)(i), $\bar{X} \sim N(40, 0.0128)$ or its equivalent was often seen but many candidates thought that $\bar{X} \sim N\left(40, \frac{0.8}{\sqrt{50}}\right)$, even though $\sigma = \frac{0.8}{\sqrt{50}}$ was correctly used in later calculations. Part (a)(ii) was often answered very well with 5% being the most popular answer seen. Some candidates unfortunately thought that this equated to 0.5, rather than the correct response of 0.05, whilst others simply described what they thought was a Type I error without giving a value for P (Type I error), as requested. This latter response gained no credit.

In part (b), most candidates realised from looking at the given hypotheses that a 2-tailed test was being considered. Many candidates used a diagram with $z = \pm 1.96$ to demonstrate the region required, but then failed to continue the process and consider $40 \pm$

$1.96 \frac{\sigma}{\sqrt{n}}$ to obtain $[39.78, 40.22]$ as the acceptance region asked for in the question. Some candidates who followed the correct method throughout lost credit for not writing the answer correct to two decimal places.

Part (c)(i) produced mixed and often confusing answers from candidates. 'Accepting H_0 when H_0 incorrect' was the expected response here. Part (c)(ii) produced some excellent solutions but only from the most able candidates. The vast majority of candidates seemed to have very little or no idea of what was expected of them. Although this part of the question covered probably the most difficult concept in the unit, it was disappointing to see the lack of understanding apparent in candidates. Candidates who used some type of

diagrammatic representation when finding the probability of a Type II error usually gained some credit for their efforts.

Q8.

This question was extremely well answered. In particular, in part (a)(ii), the majority of candidates correctly calculated a negative test statistic and compared it with a negative critical value. Part (b) was expected to offer a substantial challenge. However many candidates obtained full marks.

Q9.

Parts (a) and (b) were answered well. In particular, the great majority of answers showed correct signs for both test statistics and critical values. Part (c) was intended to refer to the nature of the tests. However, many candidates referred to the particular application and this was accepted. For example, in part (c)(i) the expected answer was “neither, since both tests have a 5% risk of Type 1 error” but answers such as “(b) - a Type 1 error cannot have been made in (a) because H_0 was accepted” were allowed.



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